

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1791

Birchwood 138 kV
June 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1791. The Interconnection Customer proposes a 79.7 MW solar photovoltaic facility interconnecting at the Birchwood 138 kV substation in the Virginia area. The request was received on June 22, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 631 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$75,024,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BLAKE / INGLESIDE 138kV	NAUVOO / BIRCHWOOD 345kV	198.3	242	11.1
INGLSIDE / COMORN DP 138kV line	NAUVOO / BIRCHWOOD 345kV	197.4	245	11.1
S KIRKLAND / BIRCHWOOD 138kV	BIRCHWOOD / NAUVOO 345kV L	188.8	279	11.1
COMORN DP / SOUTH KIRKLAD 138kV	NAUVOO / BIRCHWOOD 345kV Line	173.1	292	11.1
RYCEVILLE / BLAKE 138kV	BLAKE SUBSTATION / JAFFREY 138kV line	127.0	244	6.1
RYCEVILLE / SMILEY 138kV	BLAKE / JAFFREY 138kV	125.1	242	6.1
W MONTGOMERY / SMILEY 138kV line	BLAKE / JAFFREY 138kV	120.6	236	6.1
KEARNS / WEST MONTGOMERY 138kV	BLAKE / JAFFREY 138kV line	115.3	231	6.1
CONTRA COSTA / BIRCHWOOD 138kV	BIRCHWOOD / NAUVOO 345kV Line	111.7	537	19.9
KEARNS / RATTLESNAKE CRK WIND PROJECT 138kV	BLAKE / JAFFREY 138kV	110.1	226	6.1
BLAKE / JAFFREY 138kV	BLAKE / RYCEVILLE 138kV	105.1	212	8.7
FREDERICKSBURG / POSSUM PT 345kV	NAUVOO / BIRCHWOOD 345kV	102.3	1,600	6.4
MILLER / CONTRA COSTA 138kV line	MILLER / CONTRA COSTA 138kV	101.7	693	13.2
MILLER / CONTRA COSTA 138kV Line 2	MILLER / CONTRA COSTA 138kV	101.7	693	13.2
BIRCHWOOD / NAUVOO 345kV	DUNN SIDING / FREDERICKSBURG 345kV	101.7	2,814	22.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
NAUVOO / CITY WEST DIESEL PLANT 345kV L	NAUVOO / ROBERTS SW STA 345kV	100.3	1,910	10.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BLAKE / INGLESIDE 138kV	Rebuild line with higher-capacity conductor	\$7,972,000	100% (\$7,972,000)
INGLSIDE / COMORN DP 138kV line	Rebuild line with higher-capacity conductor	\$6,774,000	100% (\$6,774,000)
S KIRKLAND / BIRCHWOOD 138kV	Rebuild line with higher-capacity conductor	\$7,204,000	100% (\$7,204,000)
S KIRKLAND / COMORN DP 138kV	Rebuild line with higher-capacity conductor	\$2,212,000	100% (\$2,212,000)
RYCEILLE / BLAKE 138kV	Rebuild line with higher-capacity conductor	\$10,256,000	100% (\$10,256,000)
RYCEVILLE / SMILEY 138kV	Rebuild line with higher-capacity conductor	\$3,957,000	100% (\$3,957,000)
W MONTGOMERY / SMILEY 138kV line	Rebuild line with higher-capacity conductor	\$3,077,000	100% (\$3,077,000)
KEARNS / WEST MONTGOMERY 138kV	Rebuild line with higher-capacity conductor	\$1,575,000	100% (\$1,575,000)
CONTRA COSTA / BIRCHWOOD 138kV	Rebuild line with higher-capacity conductor	\$17,370,000	100% (\$17,370,000)
KEARNS / RATTLESNAKE CRK WIND PROJECT SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$5,998,000	100% (\$5,998,000)
BLAKE / JAFFREY 138kV	Reconductor line	\$5,344,000	100% (\$5,344,000)
FREDERICKSBURG / POSSUM PT 345kV L	Reconductor line	\$33,239,000	1% (\$217,000)
MILLER / CONTRA COSTA 138kV line	Reconductor line	\$4,949,000	15% (\$741,000)
MILLER / CONTRA COSTA 138kV Line 2	Reconductor line	\$11,444,000	15% (\$1,714,000)
NAUVOO SUBSTATION / BIRCHWOOD 345kV Line	Reconductor line	\$40,041,000	1% (\$446,000)
CITY W DIESEL PLANT / NAUVOO 345kV	Reconductor line	\$18,477,000	1% (\$167,000)
	Total allocated to project		\$75,024,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.